

DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning**



INTERNSHIP REPORT

ON

“VAPT (Vulnerability Assessment and Penetration Testing) Reporting Portal”

Submitted in partial fulfillment for the award of degree (21INT82)

**BACHELOR OF ENGINEERING IN
Department of Artificial Intelligence and Machine Learning**

Submitted by:

Bhargavi K (1DS21AI009)

Conducted at



**CyRAACS(Cyber Risk Advisory and
Consulting Services)**

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



CERTIFICATE

This is to certify that the Internship titled “**VAPT (Vulnerability Assessment and Penetration Testing) Reporting Portal**” carried out by **Bhargavi K [1DS21AI009]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)


Sriram Kumar Govindaraj

Signature of Reporting Head
Mr. Sriram Kumar Govindaraj
VP-Head of Product Engineering
CyRAACS

Signature of Internship Guide
Prof. Anupama V P
Dept. of AIML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Bhargavi K**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date: 11/04/2025

Place: Bangalore

USN: 1DS21AI009

NAME: Bhargavi K

OFFER LETTER



10 February 2025

Bhargavi K

Dear Bhargavi,

Sub: Internship with CyRAACS

We are pleased to offer you Internship at CYRAAC SERVICES PRIVATE LIMITED so as to facilitate you to acquire requisite degree of proficiency, competency and work knowledge.

Joining Date: 17 February 2025

Duration: 6 months

Location: Bangalore

Business Line: Product Development

Stipend: INR 10,000 only per month

You will be reporting to **Mr. Murari Shanker** and will be briefed on your Project when you join.

Your current engagement is only in the capacity of an Intern and shall not be construed to be an employment. Moreover, the current training shall not entitle you to claim permanent employment with either CYRAAC SERVICES PRIVATE LIMITED or any of the customers you may be working with for a project. Based on your performance during this internship, you may be offered a position at CYRAAC Services Private Limited.

During your internship with Company, you may have access to trade secrets and confidential business information belonging to the Company. By accepting this offer of internship, you acknowledge that you will keep all of this information strictly confidential, and refrain from using it for your own purposes or from disclosing it to anyone outside the Company. In addition, you agree that, upon conclusion of your internship, you will immediately return to the Company all of its property, equipment, and documents, including electronically stored information.

By accepting this offer, you agree that throughout your internship, you will observe all policies and practices governing the conduct of our business and employees, including our policies prohibiting discrimination and harassment.

Looking forward to having you with us, and hope your association with the Company will be successful and rewarding. If you have any questions, please contact do not hesitate to contact me.

Please indicate your acceptance of this offer by signing below.

Yours faithfully,

For **CYRAAC SERVICES PRIVATE LIMITED**

Jyothi Mehta

Head – Human Resources

Signed by Jyothi Mehta
Date: 2025.02.10
14:22:17

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Sriram Kumar Govindaraj**, for guiding me during the period of internship.

We express our deep and profound gratitude to our Internship Guide, **Prof. Anupama V P**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Bhargavi K [1DS21AI009]

TABLE OF CONTENTS

Sl no	Description	Page no
1	INTRODUCTION	1-3
2	DESCRIPTION OF ORGANIZATION	4-7
3	SCOPE OF WORK	8-9
4	LEARNING OBJECTIVES AS GIVEN BY COMPANY	10 -11
5	CHALLENGES AND SOLUTIONS	12-15
6	ACHIEVEMENTS AND CONTRIBUTION	16-31
7	REFLECTION AND ANALYSIS	32
8	RECOMMENDATIONS	33
	CONCLUSION	34
	REFERENCES	35

CHAPTER 1

INTRODUCTION

1.1 Introduction to CyberSecurity



In today's digital world, **cybersecurity** plays a vital role in protecting our data, systems, and online activities from unauthorized access, cyberattacks, and data breaches. As more organizations move their operations online, the risk of cyber threats has increased significantly. Cybersecurity involves a set of practices and technologies used to protect computers, networks, servers, mobile devices, and data from malicious attacks.

Cybersecurity is not just about installing antivirus software—it includes a **wide range of services and measures**, such as firewalls, encryption, user authentication, regular system updates, and security awareness training. These help to build strong defenses against cybercriminals and protect sensitive business and personal information.

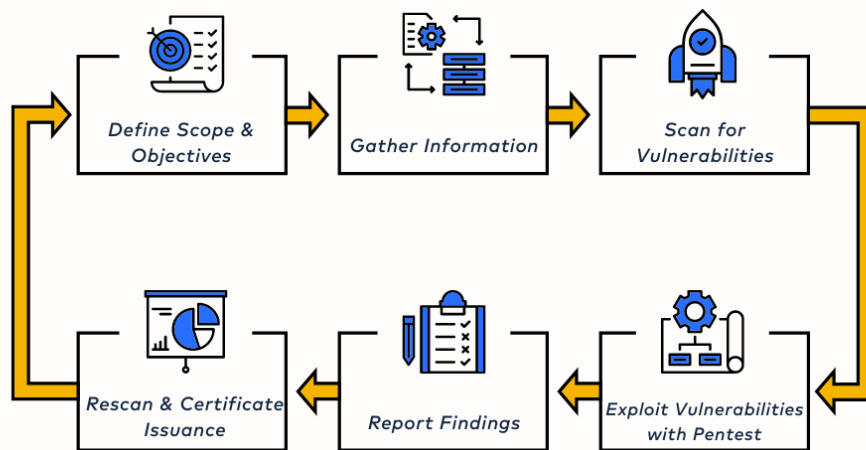
Why Cybersecurity is Important

- **Data Protection:** Protects confidential and sensitive data from theft or loss.
- **Business Continuity:** Ensures that business operations are not interrupted due to cyberattacks.
- **Compliance:** Helps organizations meet legal and regulatory requirements.
- **Reputation Management:** Prevents reputational damage caused by data leaks or attacks.

Organizations today rely on **cybersecurity service providers** to help them build strong security programs and manage ongoing risks. These services may include **risk assessments, security audits, penetration testing, incident response, and compliance support**.

1.2 VAPT (Vulnerability Assessment and Penetration Testing)

VULNERABILITY ASSESSMENT & PENETRATION TESTING



VAPT stands for **Vulnerability Assessment and Penetration Testing**. It is a process used to identify and fix weaknesses in a system's security. Although both are related, they serve different purposes:

- **Vulnerability Assessment** is the process of scanning systems, networks, and applications to detect known vulnerabilities or flaws. These flaws could be outdated software, weak passwords, misconfigurations, or missing patches.
- **Penetration Testing (Pen Testing)** goes a step further by simulating real-world cyberattacks to exploit these vulnerabilities. Ethical hackers try to break into the system to understand how an actual attacker might cause harm.
- Together, VAPT helps organizations:
 - Discover and prioritize vulnerabilities.
 - Get actionable insights for fixing issues.
 - Strengthen their security posture.
 - Prepare for real-life attack scenarios.

VAPT is a crucial service for companies that want to stay proactive against cyber threats. It is often performed regularly or after major changes in IT infrastructure, such as system upgrades or application deployments.

1.3 GRC (Governance, Risk, and Compliance)



GRC stands for **Governance, Risk, and Compliance**. It is a framework that helps organizations manage their operations while staying compliant with laws, reducing risks, and maintaining internal control.

1. Governance:

Governance refers to how an organization makes decisions and implements policies to meet its goals. In cybersecurity, governance ensures that security strategies align with business objectives.

2. Risk Management:

Risk management involves identifying, assessing, and reducing risks that could affect business operations. This includes cybersecurity risks like data breaches, ransomware, insider threats, etc.

3. Compliance:

Compliance is about following rules, laws, and industry standards such as ISO 27001, GDPR, or HIPAA. It ensures the organization is meeting regulatory and legal requirements related to data and privacy.

A GRC platform helps simplify and automate these tasks. It allows businesses to:

- Monitor risk levels.
- Conduct security assessments.
- Manage incidents and controls.
- Track compliance requirements.
- Generate reports for audits.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

CyRAACS – Cyber Risk Advisory and Consulting Services



2.1 Introduction

CyRAACS (Cyber Risk Advisory and Consulting Services) is a leading cybersecurity firm specializing in comprehensive solutions for organizations seeking to enhance their security posture and ensure compliance with industry standards. Founded by industry veterans Suresh Iyer and Murari Shanker, CyRAACS focuses on tailoring and integrating solutions seamlessly into client environments, allowing them to concentrate on their core business operations.

Services Offered:

- **Governance and Compliance:** Assistance with frameworks, assessments, implementations, audits, and third-party risk management.
- **Consulting Expertise:** Providing expert advice and strategies for robust security, compliance, and data protection.
- **Technical Fortification:** Conducting assessments, vulnerability testing, code reviews, and advanced threat analysis.
- **FinSec Solutions:** Implementing security protocols, encryption, and proactive threat identification and response.
- **Cloud Security Assurance:** Addressing security challenges and minimizing risks in public cloud environments.

Growth:



Key Value Propositions:

- **Consulting Done Right:** Combining technology, security, and deep industry experience to offer unique, tailored approaches rather than one-size-fits-all solutions.
- **Framework & Methodology:** Utilizing comprehensive compliance frameworks aligned with industry standards, employing flexible methodologies and tools to meet client requirements.
- **Technical Proficiency:** Conducting assessments with a comprehensive library of control standards, ensuring accurate and repeatable results, and providing continuous monitoring and compliance for information security sustenance.
- **Insights and Actionable Outcomes:** Delivering a comprehensive view of the security landscape with intelligent and actionable dashboards.

Innovative 5D Working Process:

CyRAACS employs a structured 5D methodology to deliver its services:

- **Discover:** Gathering information and assessing project readiness.
- **Define:** Establishing scope and conducting gap assessments.
- **Design:** Developing frameworks and providing remediation support.
- **Deliver:** Validating controls and conducting simulation exercises.
- **Drive:** Ensuring ongoing sustenance and continuous improvement.

Career Opportunities and Culture:

CyRAACS fosters a high-performance culture emphasizing customer delight and integrity. The company offers a stimulating work environment with provisions for self-learning and growth, promoting transparency, collaboration, and opportunity.

Mission: To empower organizations with secure, scalable, and sustainable cybersecurity strategies.

Vision: To be the most trusted name in the cybersecurity consulting space by delivering innovative solutions that are simple, secure, and sustainable.

Awards and Recognitions:

- Awarded Best Cyber Security Consulting Company of the Year – CISO Leadership Awards 2019.

Recognized as the Best Cyber Security Company of the Year at Fintech Indian Summit & Awards (2022).

Awarded as Cyber Security Innovator of the Year at the 6th Edition CISO Conclave & Awards (2023).

2.2 Founders

Suresh Iyer – CEO & Co-Founder



Suresh Iyer brings over 28 years of rich experience in Information Technology, IT Security, Risk Management, and Privacy. He has held CXO roles at several global organizations including Owens Financials, Altisource, Aditya Birla Minacs, eFunds, and Bank of America. Suresh is deeply passionate about Technology and Risk Management and has provided visionary leadership across Technology Risk, Operational Risk, Information Security, IT Infrastructure, and Data Centre Technologies. Known for being an eminent speaker and panelist at various industry and security forums, he shares valuable insights drawn from his vast experience. He is a dynamic, result-oriented leader who believes that hard work and commitment are key to achieving any goal.

Murari Shanker – CTO & Co-Founder



Murari Shanker is a seasoned technology leader with 30 years of global industry experience, having worked with corporations like Concentrix, Altisource, Convergys, Mphasis, and IBM. His expertise spans Technology, Data Centers, and Information Security, and he is also a Certified Information Security Manager. Murari has hands-on experience in Mergers & Acquisitions (M&A), process efficiency, and automation, coupled with strong leadership and team management capabilities. He is known for his keen attention to detail, which has helped organizations achieve cost, time, and quality targets effectively. His unique strength lies in balancing long-term strategic value with short-term execution, ensuring client success through continuous process improvements and Six Sigma methodologies, ultimately driving maximum profitability.

2.3 Compass and VAPT Portal – A Combined Approach to GRC



CyRAACS has expanded beyond cybersecurity consulting to develop its own in-house product suite—**COMPASS**, a SaaS platform that addresses Governance, Risk, and Compliance (GRC) needs. COMPASS is built to support both **enterprises** and **service providers**, offering a centralized space to manage key security operations like **third-party risk, compliance assessments, risk evaluations, and issue tracking**. Its clean interface and modular design simplify complex cybersecurity workflows, making it easier for organizations to maintain a secure and compliant posture.

The **VAPT Portal** in the process of being integrated with **COMPASS**, is a dedicated module for **Vulnerability Assessment and Penetration Testing (VAPT)**. This portal is designed to manage the entire lifecycle of vulnerability identification, tracking, and remediation. It supports both **manual and automated testing**, provides detailed **risk dashboards**, and allows teams to collaborate efficiently during security audits or incident investigations. The VAPT module not only supports security improvements but also aligns with compliance needs by ensuring vulnerabilities are resolved and documented properly.

Together, **COMPASS and VAPT** form the **complete GRC solution** for clients. While COMPASS handles governance and risk workflows, the VAPT Portal strengthens the compliance and technical testing aspects. By bringing these capabilities under one platform, CyRAACS offers a scalable and integrated system that enhances security visibility, operational efficiency, and regulatory readiness for its clients.

CHAPTER 3

SCOPE OF WORK

The scope of work involved enhancing the internal VAPT Reporting Portal by developing and integrating both frontend and backend features. The project focused on file validation, backend logic, API integration with MongoDB, and fixing critical bugs to improve usability. These tasks aimed to strengthen the portal's functionality, ensure secure data handling, and support real-time collaboration across teams.

1. Frontend Validation and Error Handling

- **Objective:**
To enhance the user experience and data accuracy by implementing frontend validation for file uploads and input fields.
- **Scope:**
This included adding file-type validation (e.g., JSON, image files), handling incorrect inputs, and providing user feedback using notification tools like React Toastify.
- **Methodology:**
Implemented conditional checks within React components to restrict unsupported file types and displayed real-time error messages for invalid user inputs.
- **Results:**
Improved frontend reliability, reduced incorrect file uploads, and provided a smoother user experience through responsive error notifications.

2. Backend Logic Implementation

- **Objective:**
To strengthen the backend structure for secure data handling, request processing, and response generation.
- **Scope:**
This involved developing and refining API endpoints using FastAPI to support file handling, data validation, and structured JSON responses.
- **Methodology:**
Designed robust backend logic to manage file data, ensured schema consistency through Pydantic models, and integrated endpoint responses with frontend components.

- **Results:**

Achieved efficient backend functionality with reduced error rates and seamless communication between client and server.

3. Testing and Bug Fixes

- **Objective:**

To identify and resolve existing issues in frontend-backend communication and application behavior during development.

- **Scope:**

Covered form synchronization bugs, environment-based test errors, and UI inconsistencies.

- **Methodology:**

Conducted testing using Postman for backend endpoints, used browser dev tools for frontend debugging, and collaborated with team members to isolate and fix bugs.

- **Results:**

Enhanced application stability, reduced form handling errors, and ensured smoother development cycles through systematic debugging.

4. Integration of MongoDB with FastAPI

- **Objective:**

To integrate a cloud-based database (MongoDB Atlas) with FastAPI for efficient storage, retrieval, and manipulation of portal data.

- **Scope:**

Involved setting up secure database connections, implementing CRUD operations, and managing schema design to support application workflows.

- **Methodology:**

Utilized MongoDB client libraries to establish a secure connection to Atlas, designed collections and indexes, and linked backend endpoints to perform real-time operations.

- **Results:**

Enabled persistent and scalable data handling, improved API performance, and supported secure data storage for VAPT reports and user actions.

CHAPTER 4

LEARNING OBJECTIVES

1. Strengthen Full-Stack Development Skills

The primary learning objective of this internship was to enhance my understanding and practical application of full-stack development using modern technologies. Working on a live internal project, the VAPT Reporting Portal, allowed me to implement features both on the frontend and backend. On the frontend, I applied React.js concepts such as `useState`, `useEffect`, `useRef`, and `useContext` to manage component state, fetch data, and handle user input. On the backend, I worked extensively with FastAPI, handling routing, request parsing, and response modeling. I also integrated backend services with MongoDB Atlas, performing CRUD operations and indexing. This end-to-end involvement helped solidify my grasp of how user interfaces interact with APIs and how data flows between the client and the server.

2. Understand Cybersecurity Practices and VAPT Processes

A significant objective was to familiarize myself with core cybersecurity concepts, particularly around Vulnerability Assessment and Penetration Testing (VAPT). Through direct involvement in improving the VAPT Reporting Portal, I gained insights into how organizations assess and report security vulnerabilities, how findings are categorized, and how remediation recommendations are tracked. Additionally, I learned how VAPT ties into broader Governance, Risk, and Compliance (GRC) efforts, especially through CyRAACS' product Compass, which is set to integrate the VAPT portal for a unified cybersecurity management solution. This hands-on exposure helped me understand the practical application of cybersecurity strategies in protecting infrastructure and ensuring regulatory compliance.

3. Improve Collaboration and Version Control Proficiency

Another key objective was to improve my ability to work in collaborative development environments using Git-based version control systems. Throughout the internship, I used GitLab for issue tracking, staging changes, creating merge requests, and pushing updates to development branches. I followed structured workflows, regularly testing backend changes locally and syncing them with DEV environments hosted via GitHub. This helped me understand the importance of clean commit histories, proper branching strategies, and

merge conflict resolution. It also taught me how teams manage real-time updates across environments without affecting production, an essential skill for working in software teams.

4. Develop Robust Error Handling and File Management Skills

Ensuring secure and error-free user experiences was another important learning objective. I implemented form validation mechanisms on the frontend to control user inputs and prevent improper data submission. I also worked on restricting file uploads to specific formats (such as images and JSON), thereby avoiding potential vulnerabilities. This involved using React Toastify for real-time error alerts and FastAPI backend logic for server-side validations. These tasks helped me understand how critical proper input validation and file handling are to prevent issues like broken APIs, insecure uploads, or inconsistent data—especially in cybersecurity tools dealing with sensitive reports.

5. Gain Exposure to Real-World Project Management

Lastly, the internship provided me with an opportunity to observe and participate in real-world project management workflows. I was included in planning discussions, feature reviews, and internal testing phases. I worked closely with mentors and team members to fix bugs, update UI components, and refine user interactions based on feedback. I maintained proper folder structures, consistent naming conventions, and development notes to ensure code clarity and maintainability. Through this experience, I gained a deep appreciation for agile development practices and learned how even minor UI/UX changes can significantly improve product usability in professional environments.

CHAPTER 5

CHALLENGES AND SOULTION

1. Static Date Display for Issues

Challenges:

In the initial design of the VAPT Reporting Portal, the date field for issues was hardcoded or manually entered, which resulted in errors and inconsistencies. This static handling of dates caused confusion for users as it did not accurately reflect the actual creation or update times of entries. It also limited the reliability of audit logs and historical tracking of issue progress.

Solutions:

To overcome this, backend logic was updated to dynamically fetch the current system date and display it automatically when a new issue was created or modified. This ensured that all issues reflected accurate and real-time timestamps, improving data reliability and traceability within the portal.

2. Severity & CVSS Score Not Editable

Challenges:

Originally, the fields representing severity level and CVSS (Common Vulnerability Scoring System) score were fixed and non-editable, which restricted the flexibility needed for different vulnerability scenarios. This limitation hindered analysts from tailoring the reports based on updated vulnerability impacts or reassessments.

Solutions:

Both fields were made customizable through backend support and dynamic form components on the frontend. This allowed users to manually adjust the severity and CVSS score based on evolving threat analysis or updated vulnerability intelligence, thereby enhancing the accuracy and usefulness of each report.

3. Issues Not Saving Automatically

Challenges:

Users often forgot to manually save their inputs when documenting findings or editing issue fields, which led to frequent data loss. This was particularly problematic during multitasking or lengthy analysis sessions, where unsaved work could be lost accidentally.

Solutions:

An auto-save feature was integrated into the portal using JavaScript triggers and backend API calls to save form data periodically or when fields were modified. This greatly improved user experience by ensuring that progress was retained without requiring constant manual intervention, boosting productivity and reducing frustration.

5. No File Validation for Uploads

Challenges:

Users were able to upload files of any type without restriction. This posed a major security risk, as uploading executable or irrelevant files could compromise system security or corrupt the database. Moreover, there was no error feedback if an incorrect file type was uploaded.

Solutions:

File-type validation was added both at the frontend and backend. Only specific formats like .jpeg, .jpg, and .png were allowed. Additionally, real-time error notifications were introduced using React Toastify, helping users instantly understand and correct their mistakes. This reduced invalid submissions and strengthened data security.

6. File Upload Accepting All Formats

Challenges:

In addition to the lack of validation, the system was previously configured to accept all file types without checks. This introduced inconsistencies and increased the likelihood of uploading unsupported or potentially harmful files.

Solutions:

The upload logic was rewritten to restrict uploads to only accepted formats. On the backend, conditional checks were implemented using FastAPI to block unsupported file types. On the frontend, user prompts and file input restrictions guided users in selecting the right file formats, improving the system's robustness and hygiene.

7. Difficulty Learning React & FastAPI Initially

Challenges:

As someone relatively new to the tech stack, initial learning curves with React (frontend) and FastAPI (backend) were steep. Understanding component lifecycles, hooks, state management in React, and asynchronous routing in FastAPI was challenging without prior experience.

Solutions:

To overcome this, I adopted a hands-on learning approach by building mini-projects using both frameworks in isolation. This helped me gradually understand their inner workings, structure, and best practices. Team mentorship and code reviews also played a key role in accelerating my learning curve and improving code quality over time.

8. Sync Issues Between Frontend and Backend

Challenges:

Initially, API calls made from the frontend did not align with the backend routes or data formats, resulting in failed requests and inconsistent behavior. This caused delays during integration phases and made it difficult to debug issues efficiently.

Solutions:

I used Postman to test and verify backend API endpoints before integrating them with the frontend. This isolated debugging helped identify issues faster. With team collaboration, I handled asynchronous requests correctly in React and aligned the data payloads for seamless communication between the UI and the server.

9. Complex and Confusing Project Structure

Challenges:

The original folder and file organization in the project lacked clarity, with overlapping naming conventions and minimal internal documentation. This made onboarding, navigation, and debugging tedious, especially when working across multiple modules.

Solutions:

I reorganized the project files based on functional domains and added developer notes for easy reference. Naming conventions were standardized to ensure clarity and modularity. These changes significantly improved code maintainability and made the codebase more approachable for new contributors and reviewers.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

Operating System: Windows 10/11

Development Tools:

- Visual Studio Code
- Git
- Postman

Programming Languages and Frameworks:

Frontend:

- Language: JavaScript
- Framework: React.js (used for UI development and component-based structure)
- Libraries Used:
 - Axios – for HTTP requests
 - React Toastify – for real-time error notifications
 - Tailwind CSS (if used) – for quick UI styling and responsive design

Backend:

- Language: Python
- Framework: FastAPI (used for building secure, asynchronous REST APIs)

Database: MongoDB Compass (for storing reports and issue metadata)

Runtime and Package Managers:

- Node.js and npm – For managing JavaScript packages and running React dev.
- Python 3.8+ and pip – For managing backend dependencies and running FastAPI servers.

Hardware requirements:

- Processor: Intel i5 or AMD Ryzen 5 and above
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: At least 256 GB SSD
- Internet: Stable high-speed connection

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

During my internship at CyRAACS, I had the opportunity to contribute significantly to the VAPT Reporting Portal by working on a series of frontend and backend enhancements, as well as actively participating in development, testing, and deployment workflows. Each contribution not only helped improve the portal's performance and usability but also enabled me to strengthen my full-stack development skills.

1. Frontend Enhancements

One of my major responsibilities was improving the user interface to ensure a better experience for both new and returning users. I focused on making the portal more visually appealing, intuitive, and responsive.

- **UI/UX Improvement:** I redesigned certain components to ensure consistency in layout, color schemes, and element placement. The goal was to reduce visual clutter and simplify the user experience, especially during the issue submission process.
- **Image Format Validation:** To improve file input reliability, I added a validation mechanism to accept only specific image formats (JPEG, JPG, PNG). This reduced the risk of incompatible file types being uploaded, ensuring smooth image rendering and improving backend processing.
- **Real-time Feedback with Toast Messages:** To give users immediate visual cues on their actions (e.g., form submission, errors, successful uploads), I integrated toast messages using a notification library. This significantly enhanced interaction feedback, helping users stay informed about the status of their actions without navigating away from the page.

2. Backend Improvements

My backend contributions involved enhancing the logic behind issue creation and improving data handling to ensure accurate and user-customizable reporting.

- **Custom Severity and CVSS Scoring:** Previously, these fields were static. I added support for dynamic selection, enabling users to input and edit severity levels and CVSS scores based on their assessment. This flexibility made the portal more aligned with real-world VAPT reporting standards.

- **Fixing Date Display Logic:** While uploading issues, the date field used to display incorrect timestamps. I debugged the issue and implemented logic to fetch and display accurate timestamps that align with the user's local time zone.
- **Auto-Save and Dynamic Issue Creation:** I implemented a dynamic form system that allows users to create multiple custom issues in real-time. To prevent data loss during network issues or accidental reloads, I also integrated an auto-save mechanism that temporarily stores data and repopulates it when the user returns.

3. Development, Testing, and Deployment Workflow

This phase of my work revolved around ensuring that the implemented features were stable, efficient, and well-integrated across the system.

- **Full-Stack Feature Development:** I built and tested key features combining **React (frontend)**, **FastAPI (backend)**, and **MongoDB (database)**. This provided a hands-on opportunity to apply my learning in a real-world development setting.
- **Cross-Component Issue Validation:** I worked on verifying data flow and issue creation logic across frontend forms, backend APIs, and the database, ensuring smooth interaction and error-free submissions.
- **Code Optimization:** While reviewing the existing codebase, I identified unused or redundant fields/components in the UI and removed them to improve performance and reduce cognitive load for users.
- **Debugging & Git Workflow:** I regularly debugged features in the DEV environment, using browser tools, logs, and API traces. For collaboration and version control, I followed a Git-based workflow. I used **GitLab** to manage feature branches, conducted peer reviews, and staged changes before merging. Once validated, the final changes were pushed to **GitHub** for deployment.

Technology stack

The development of the VAPT Reporting Portal involved a modern and scalable technology stack that enabled full-stack web development, real-time testing, and smooth deployment processes.

Frontend Technologies

- **React.js:** A popular JavaScript library used to build dynamic and responsive user interfaces. It enabled the development of reusable components and efficient rendering, improving the overall user experience.

- **JavaScript, HTML5, CSS3:** These standard web technologies were used to structure the application layout, implement logic, and apply styling to make the interface user-friendly and visually appealing.

Backend Technologies

- **FastAPI (Python):** A high-performance web framework used to build RESTful APIs. It enabled fast development with automatic data validation, asynchronous programming, and clear routing mechanisms to handle HTTP requests efficiently.

Database

- **MongoDB Atlas:** A cloud-hosted NoSQL database that stored user data, issue reports, and other dynamic content. It provided scalability, flexibility in schema design, and efficient storage for JSON-like documents.

API Testing

- **Postman:** Used to test backend API endpoints during development. It allowed sending custom HTTP requests, validating responses, and identifying bugs before integrating with the frontend.

Version Control

- **Git (GitHub & GitLab):** Version control was managed using Git. GitLab was used during the development phase for branching, staging, and merge requests, while GitHub was used for maintaining the final deployment-ready repository.

Notification System

- **React Toastify:** A lightweight and customizable library used to display real-time toast notifications to users. It was used to show success, error, and alert messages after form submissions or actions.

HTTP Requests

- **Axios:** A promise-based HTTP client used in React for sending asynchronous requests to the FastAPI backend. It enabled easy integration with API endpoints and efficient data handling between client and server.

Development Tools

- **VS Code:** The primary code editor used for writing and organizing code. It offered useful extensions for linting, formatting, and Git integration.
- **Terminal & Git Bash:** Used for command-line operations like running servers, pushing code, and managing version control through Git commands.

Development Workflow And Process

During my internship at CyRAACS, I worked on the VAPT Reporting Portal, where I followed a structured and iterative software development process. This workflow ensured that each feature or fix was handled systematically, from problem discovery to deployment. The approach supported collaboration, code quality, and alignment with business objectives. Below are the detailed steps I followed throughout the development lifecycle:

- **Problem Identification and Solution Design**

Each task began with identifying a problem or understanding a new feature request. These were typically raised during team discussions or by proactively noticing gaps or limitations in the current implementation. Examples included user interface issues, missing validations, bugs in API behavior, or feature enhancement requirements. After identifying the core problem, I designed a solution considering user needs, technical feasibility, and simplicity. I documented the intended logic, outlined the affected components, and clarified inputs and outputs to avoid rework later in the process.

- **Code Implementation – Frontend and Backend**

After designing the solution, I began the development phase, which involved both frontend and backend components:

- **Frontend (React.js):** I created or updated UI components using React. This included modifying layouts, adding new form fields, handling state, and ensuring a consistent and responsive design. I used libraries like Formik and Yup for form handling and validation to provide a smoother user experience.
- **Backend (FastAPI):** On the backend, I developed RESTful API endpoints using FastAPI. These endpoints handled operations such as data fetching, creation, updates, and deletions. I implemented input validations and error handling, and ensured secure interactions with the MongoDB database using asynchronous calls (via Motor). I kept the code modular, with clearly defined routes and database service layers.

- **API Testing Using Postman**

Before integrating with the frontend, I tested each backend API using Postman. I verified that all endpoints responded correctly to valid inputs and handled invalid or missing data with appropriate error messages. This step ensured that the logic on the backend was functioning correctly in isolation, which made the later frontend integration smoother and helped avoid runtime bugs.

- **Testing in Localhost and DEV Environment**

After successful backend testing, I ran the full stack locally on my development machine to simulate actual user interaction. This allowed me to verify the integration between frontend and backend, validate data flow across components, and check for UI or usability issues. Once everything worked as expected locally, I deployed the changes to the DEV environment used by the team. This environment mirrored the production setup, and testing here helped identify environment-specific issues such as CORS policies, routing behavior, or performance inconsistencies.

- **Review and Approval Process**

When development and testing were complete, I pushed my feature or bug-fix branch to the remote GitLab repository. I raised a Merge Request (MR) linked to the specific issue or feature. The MR was reviewed by senior developers or team leads who checked the logic, code readability, naming conventions, and whether best practices were followed. They sometimes provided suggestions or asked for changes, which I implemented before the code was finally approved. This review process ensured code quality and team-wide standards were maintained.

- **Staging and Merging Using GitLab**

After receiving approval:

- I staged all validated changes and ensured the branch was up to date with the latest development branch to avoid conflicts.

- I submitted a final Merge Request with comments explaining the changes and references to issue numbers.
- The QA or tech lead did a final validation of the merged code in the staging environment.
- Once verified, the MR was merged into the main development branch.

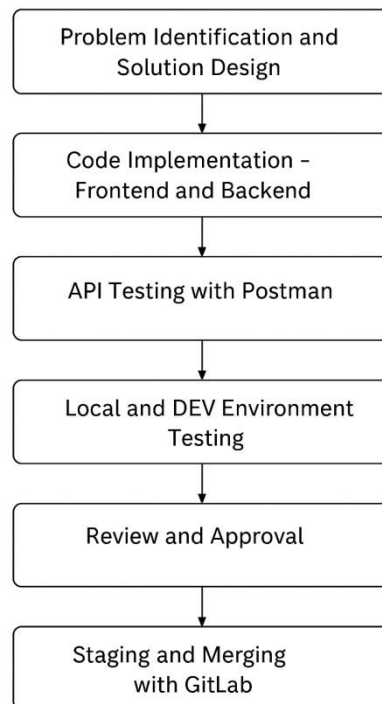


Fig 6.1: Development and Testing

Frontend:

1. Improved UI/UX to make it visually appealing and easy to understand for new users:

- Unnecessary UI components were removed to enhance clarity and performance.
- Themes and interactive features were added to create a more engaging user experience.
- Core UI elements were improved for better accessibility, feedback, and responsiveness.

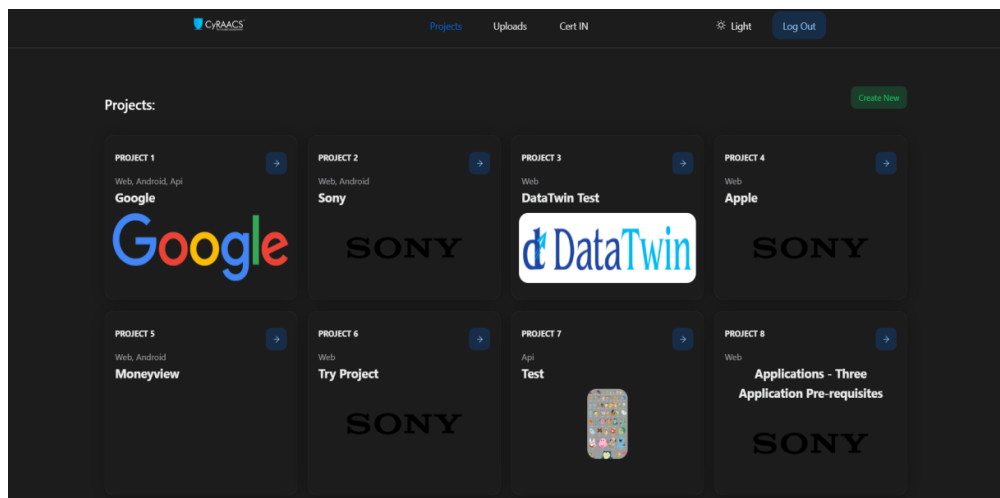


Fig 6.2: VAPT Reporting Portal

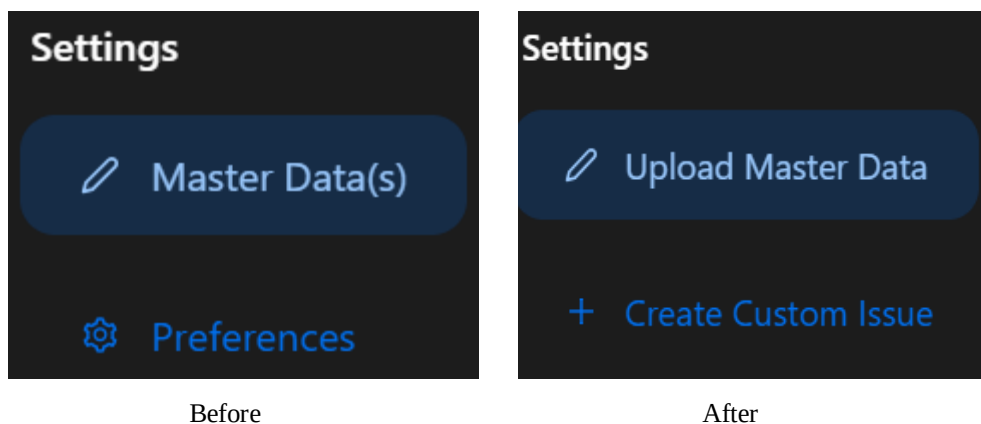


Fig 6.3: Removed Preferences and added custom issue creation

1. Added validation to accept only JPEG, JPG, PNG image formats for image uploads. Toast messages to only accept specific formats of files for images and not any Other formats of files gives error toast i.e., only JPEG, JPG and PNG rest throws an error as invalid file type

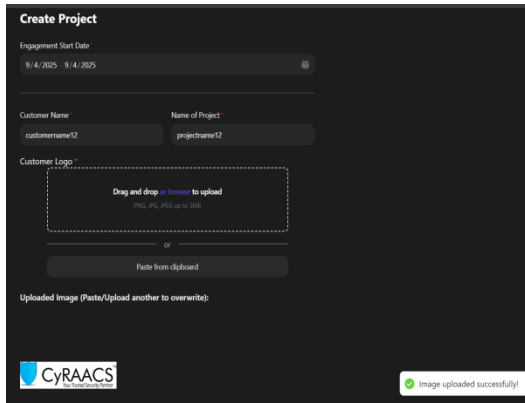


Fig 6.3: Success Toast

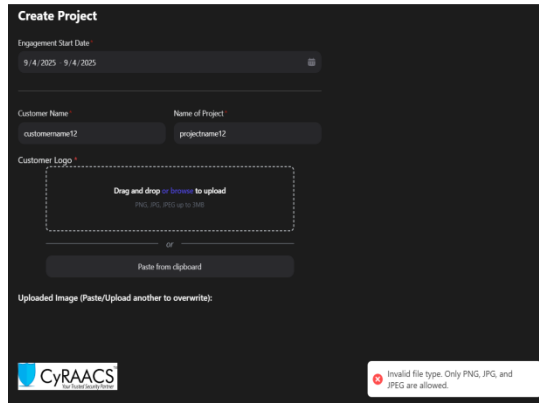
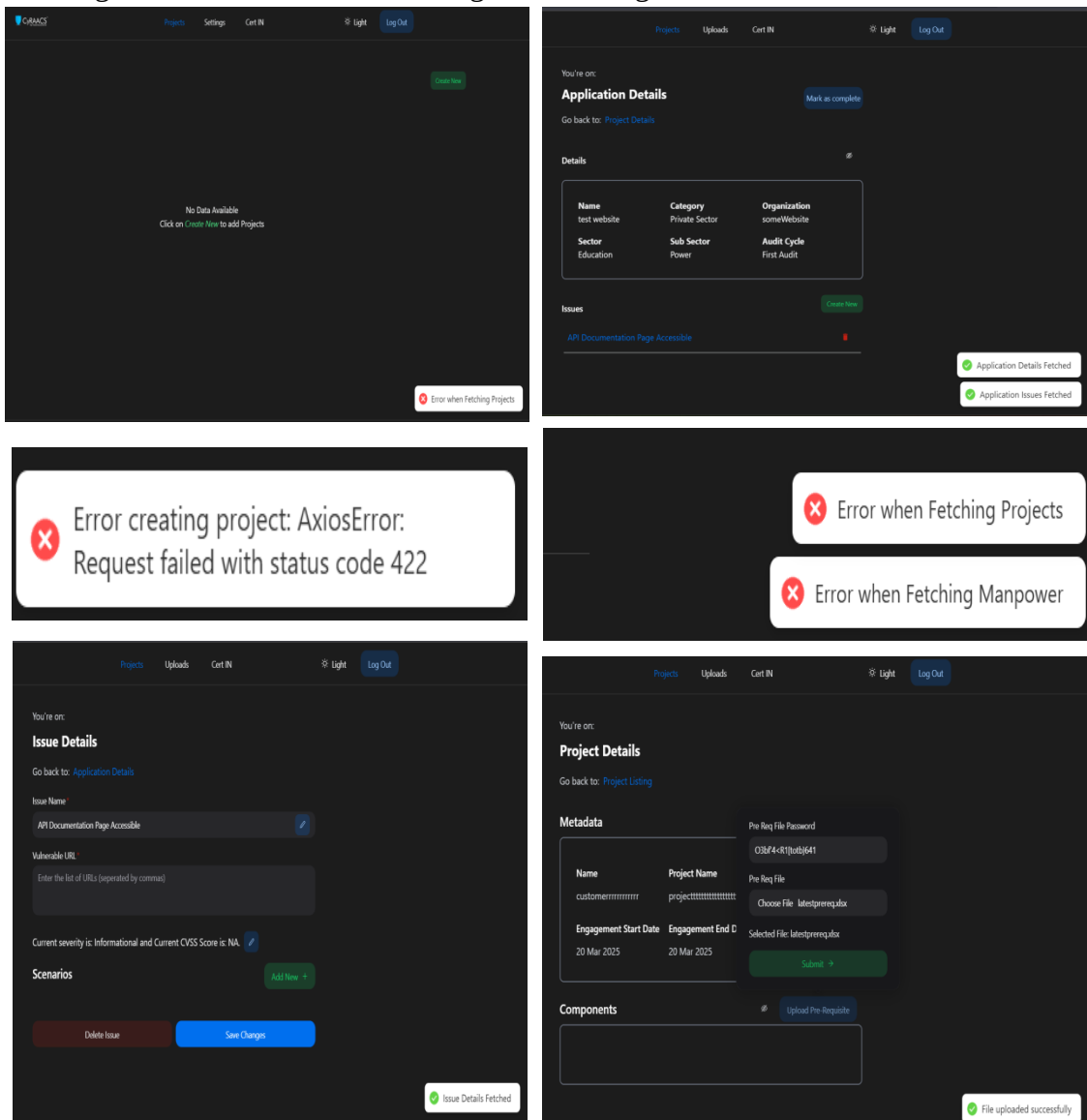


Fig 6.4: Error Toast

2. Integrated real-time feedback using toast messages.



Figs 6.5: Toast messages using Toastify from React

3. Made severity and CVSS score customizable by users.

Before the severity and CVSS were already predefined and set based on parameters but it is now set to be customizable, severity as dropdown and CVSS score as integer between 1 to 10 upto 1 decimal point only.

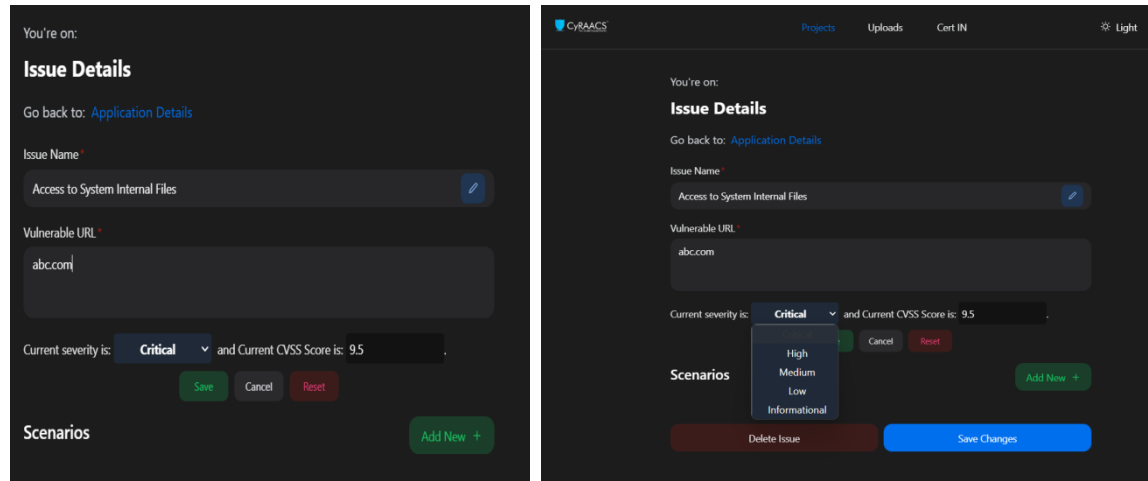


Fig 6.6: Customizable Severity and CVSS score

4. Fixed issue upload logic for accurate date display.

Previously, the timestamps on the portal were based on the time the website was opened, which caused inaccurate display. This was improved to capture the exact timestamp when a file is uploaded and submitted. The accurate upload time is now stored in the MongoDB database and the most recent timestamp is correctly displayed on the frontend.

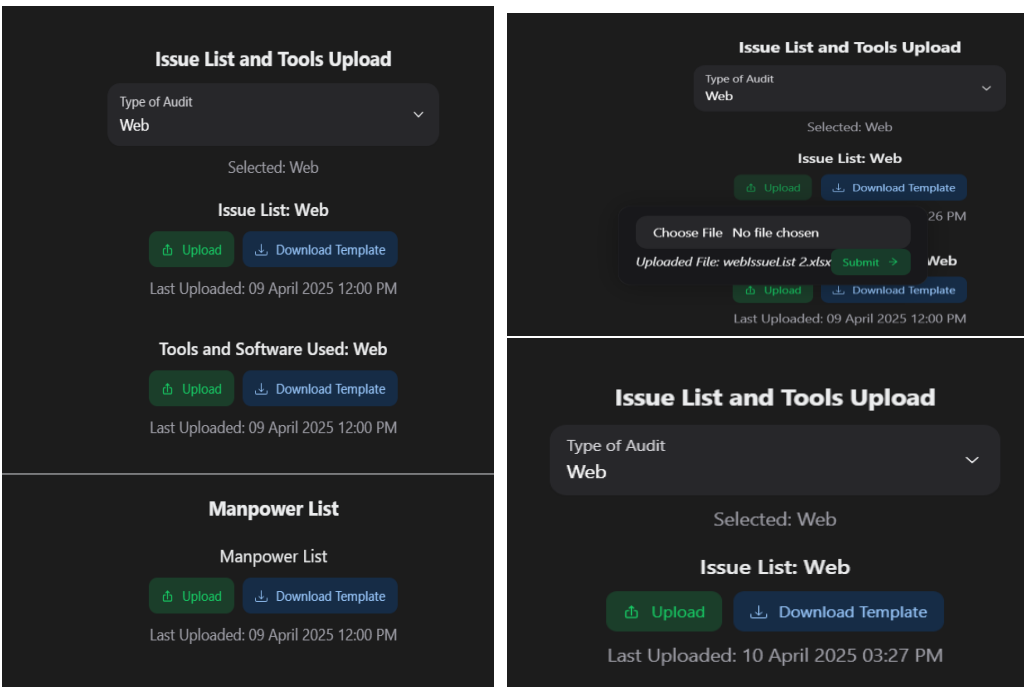


Fig 6.7: Updated Timestamps autosaving to database

6. Enabled dynamic issue creation with auto-save support

The existing “Add Issue” functionality fetches a limited set of predefined issues stored in the database. The newly implemented customized issue addition allows users to create and save a completely new issue when it doesn't already exist in the system. This new issue is auto-saved in the database and immediately reflected in the “Add Issue” component, making it available for selection and further processing based on the required parameters.

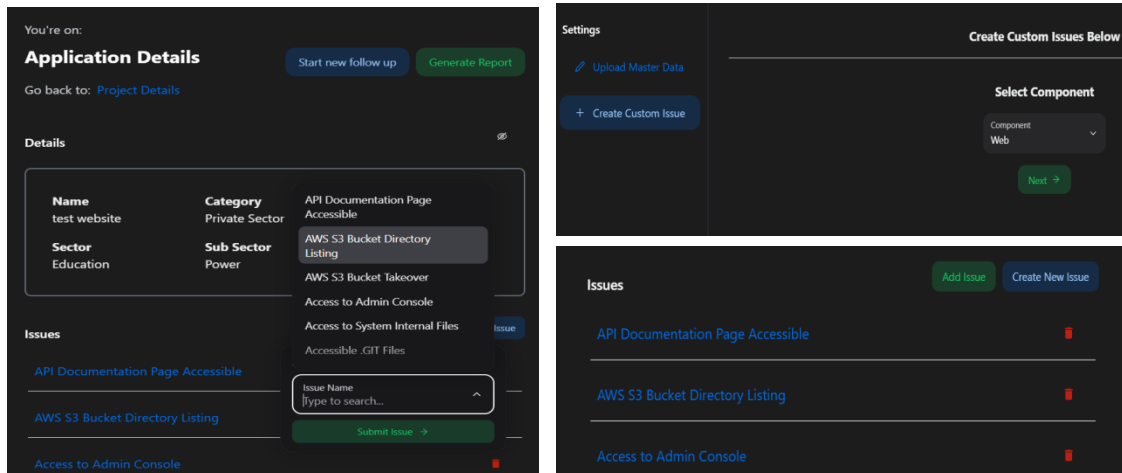


Fig 6.8: Create Custom Issue Component with existing Add Issue component

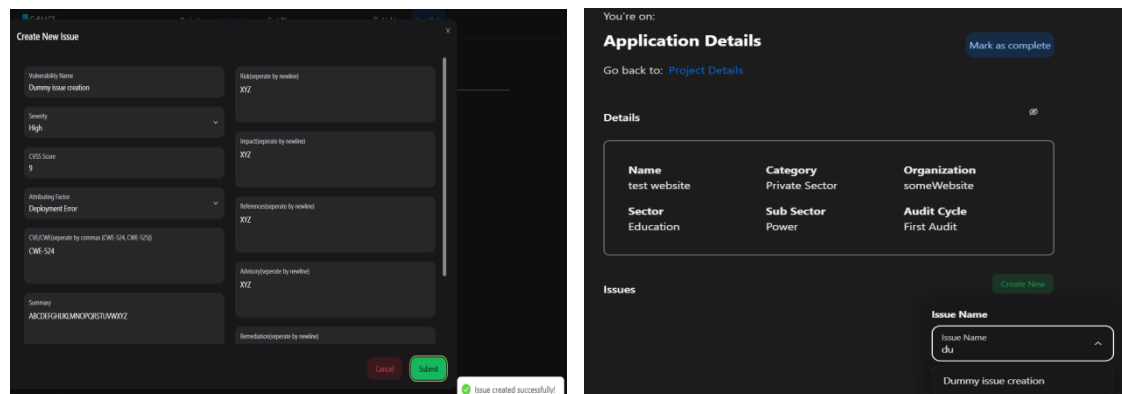


Fig 6.9: Autosaving of the Custom Issue and accessed through Add issue

7. Staging and Merging Changes in Gitlab

After coding, and evaluation the code is staged and merged with main branch to commit changes into the main development website

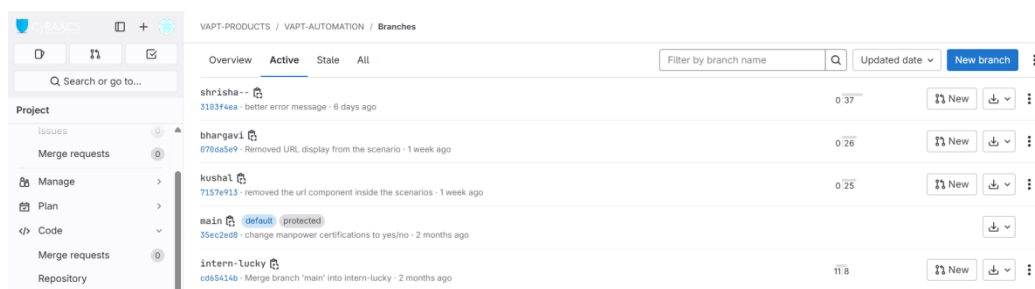


Fig 7.1: Merge code into main environment

8. Search Component, Component based Filter & UI Enhancements

To enhance the user experience and improve project accessibility, I implemented a search functionality that allows users to search projects dynamically. Additionally, I developed a component-based filter bar to sort projects by component type. UI improvements were also made by applying better spacing, font hierarchy, and styling for a cleaner, more user-friendly layout.

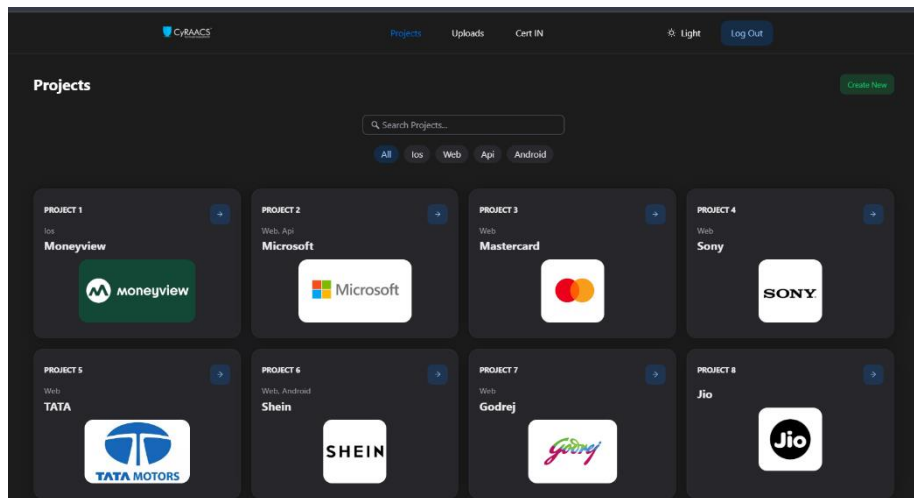


Fig 8.1: UI main page of the project

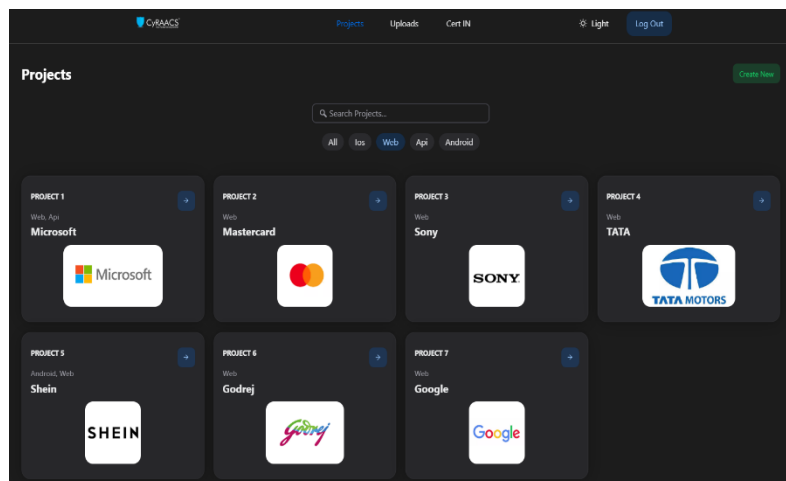


Fig 8.2: Filter based on component “web”

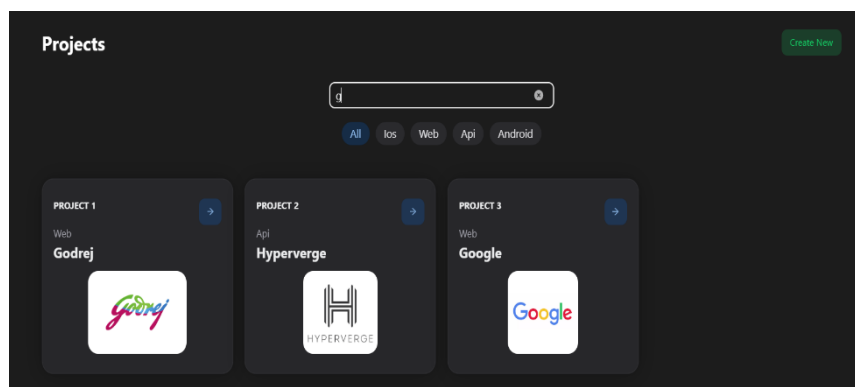


Fig 8.3: Search for a project by component or name

9. Follow-Up Report Confirmation Popup Dialog box

I introduced a confirmation dialog box that appears when initiating a Follow-Up Report. This step prevents accidental triggers of the irreversible action by requiring explicit user confirmation. It improves the workflow safety and ensures critical actions are acknowledged.

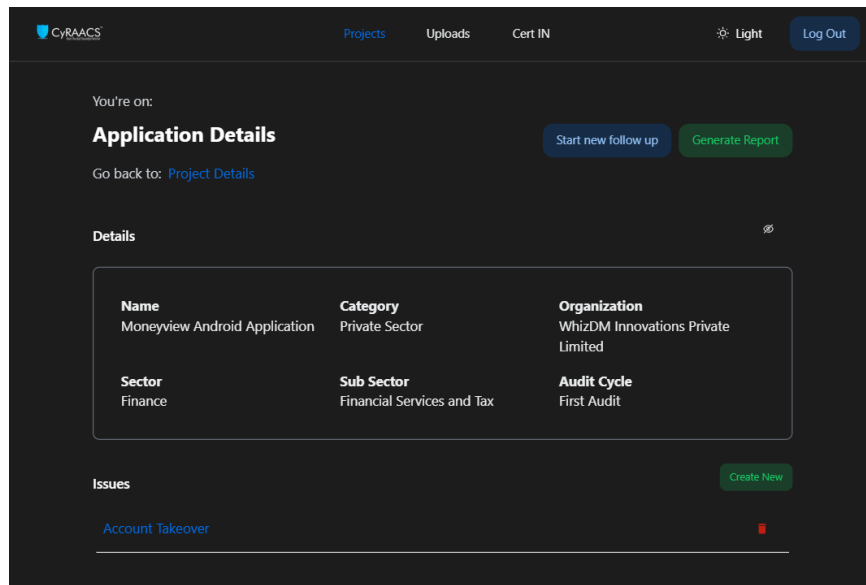


Fig 9.1: Start follow up button for starting the validation(final) report

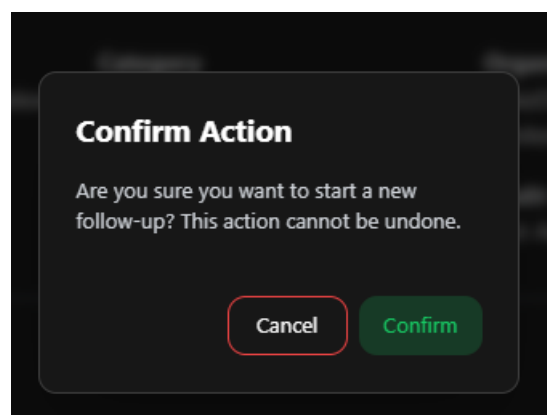


Fig 9.2: Start follow up button confirmation popup dialog box

10. Informational Severity type Removal

As per updated reporting guidelines, "Informational" severity issues are no longer required in Initial Assessment and Validation Reports. Since these issues pose no real risk or consequence (both rated 0 or NA), they make no difference to the final assessment. I implemented logic to allow the deletion of such entries directly from the UI, and this

change is reflected in the backend as well as making the reports more concise and focused only on relevant vulnerabilities.

8. Table of Vulnerability

The below table details the various severities of vulnerabilities identified as an outcome of the engagement.

SI No.	Vulnerability	Severity	Status
1	Apache Struts Remote Code Execution	INFORMATIONAL	Not Remediated
2	AWS S3 Bucket Takeover	HIGH	Not Remediated
3	Authentication Bypass using Random OTP	LOW	Not Remediated

Fig 10.1: Informational severity is displayed in the initial and validation report

8. Table of Vulnerability

The below table details the various severities of vulnerabilities identified as an outcome of the engagement.

SI No.	Vulnerability	Severity	Status
1	AWS S3 Bucket Takeover	HIGH	Not Remediated
2	Authentication Bypass using Random OTP	LOW	Not Remediated

Fig 10.2: Informational severity removed from the initial and validation report

11. Executive Summary Enhancements

To improve visual clarity and usability in the report, I added color coding to severity levels in the Executive Summary section. This enhancement helps users easily distinguish between different levels of vulnerabilities at a glance when generating or reviewing reports.

Critical – Red, High – Dark Red, Medium – Yellow, Low – Blue, Informational – Grey.

8. Table of Vulnerability

The below table details the various severities of vulnerabilities identified as an outcome of the engagement.

SI No.	Vulnerability	Severity	Status
1	Cross Domain Script Include		Not Remediated

Fig 11.1: Invisibility of the text of severity and color code

8. Table of Vulnerability

The below table details the various severities of vulnerabilities identified as an outcome of the engagement.

SI No.	Vulnerability	Severity
1	Apache Struts Remote Code Execution	CRITICAL
2	AWS S3 Bucket Takeover	HIGH
3	Authentication Bypass using Random OTP	LOW

Fig 11.2: Color code for each severity added into the report

12. Component-wise CVSS Display

Previously, the CVSS score and severity were fetched by simply retrieving the first matching issue name from the database, which often resulted in inaccurate values—especially when the same issue existed across multiple components like iOS, Web, Android, and API.

To resolve this, I implemented logic to map the CVSS score and severity based on the specific component. Now, the Issue Details view accurately reflects the correct values for each component and issue combination.

This improvement ensures precise vulnerability assessment, allowing analysts to make better-informed security decisions during reporting.

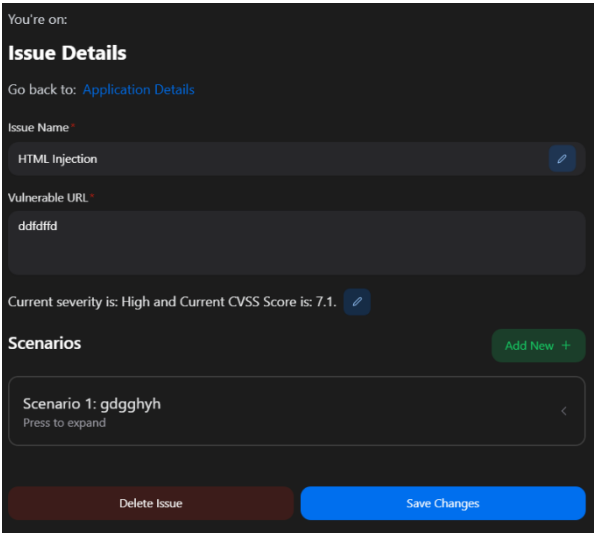


Fig 12.1: The “HTML Injection” issue for web component

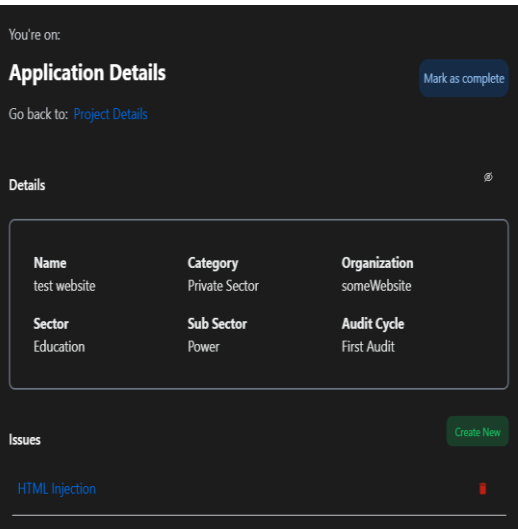


Fig 12.2: Application page with issue listed for web component

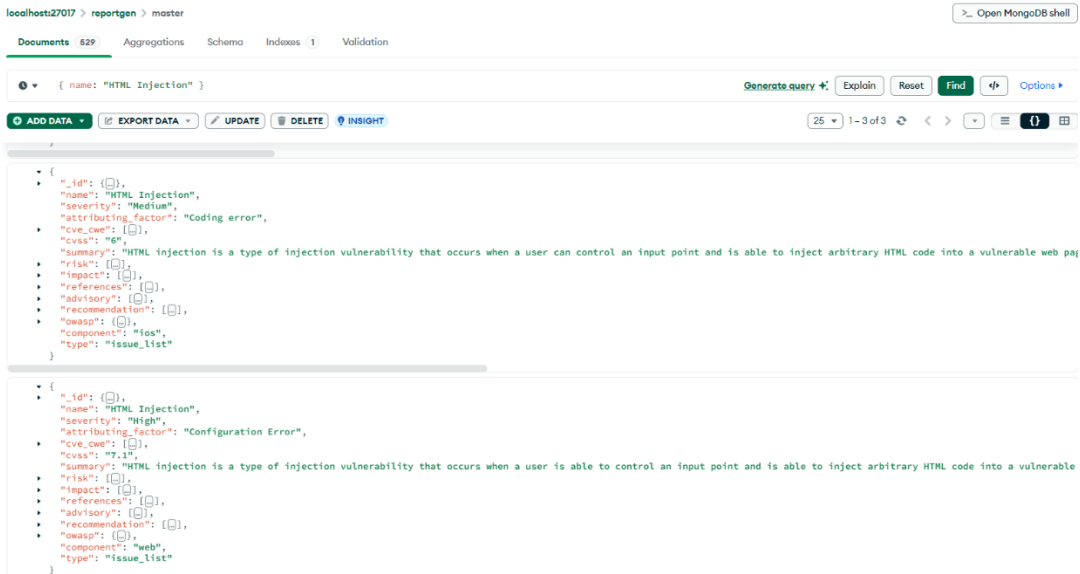


Fig 12.3 All the cvss score and severity for all components for issue “HTML Injection”

13. Bug Fixes

Several UI and data-related bugs were identified and resolved to improve consistency, clarity, and user experience:

- Removed the redundant "Component Type" field from the Project Details page—classification is now based on the uploaded pre-requisite file to ensure consistency—and cleaned up the Issue Details section by correcting spelling mistakes and eliminating duplicate "URL" labels.

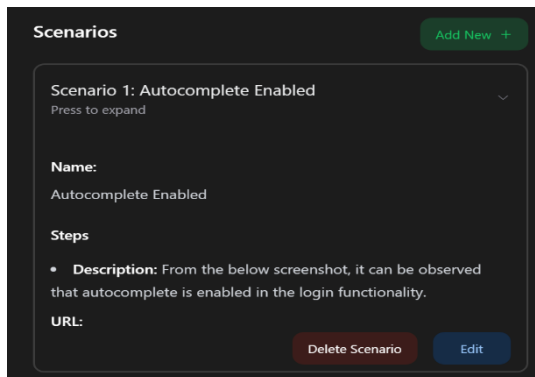


Fig 13.1: URL is unused so better to be Removed

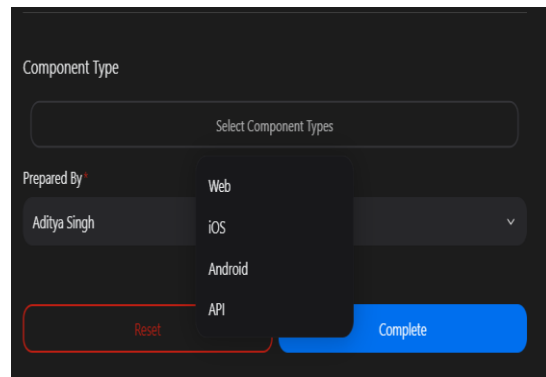


Fig 13.2: Component Type not required

- Removed the duplicated OWASP category entry ("A8 - Software and Data Integrity Failures") from the MongoDB master collection, ensuring accurate and non-redundant display in the UI.



Fig 13.3 Duplicate value of "A8 - Software and Data Integrity Failures"

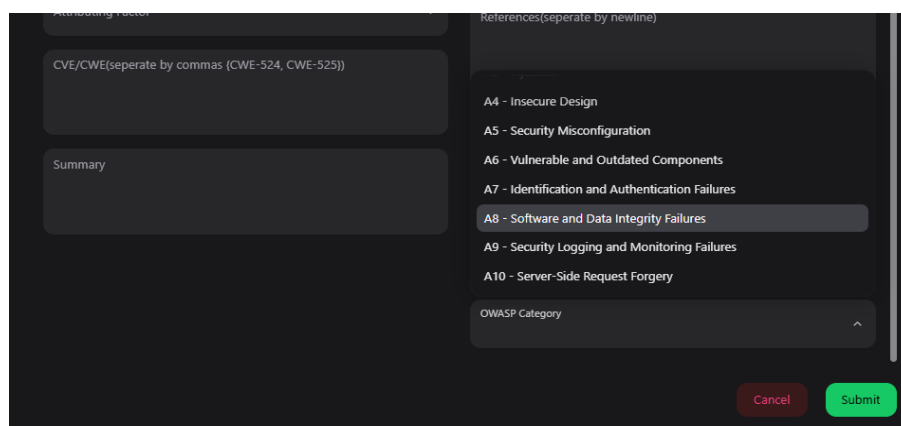


Fig 13.4 Fixed the duplicate of "A8 - Software and Data Integrity Failures"

- Fixed SPA routing and improved user flow by ensuring the pre-requisite dialog box closes automatically after submission, enabling seamless redirection to the Application Details page without requiring a manual refresh. This resolves the issue caused by overlapping isOpen states from multiple useDisclosure hooks.

The screenshot shows the 'Project Details' page. At the top, it says 'You're on: Project Details' with a link to 'Go back to: Project Listing'. Below this is the 'Metadata' section with an 'Edit' button. It contains a table with the following data:

Name	Project Name	Prepared By
Mastercard	Mastercard-Project	Pranay Sahariah

Below the table, there are two more fields: 'Engagement Start Date' (21 Apr 2025) and 'Engagement End Date' (21 Apr 2025). Under the 'Components' section, there is a 'Web' tab and a button 'Upload Pre-Requisite'. The 'Application Name' is 'test website' with a 'Go to page' button.

Fig 13.5: SPA routing of Project Details

The screenshot shows a warning dialog box titled 'Warning: Deleting Project'. It contains the text: 'You are about to delete the', 'Are you sure? This action is irreversible.', and two buttons: 'No, keep project' and 'Yes, delete'. In the background, there is a form for 'Pre Req File Password' and 'Pre Req File'.

Fig 13.6: Issue with overlapping “useDisclosure” using same functions

The screenshot shows the 'Project Details' page with the 'Pre Req File Password' and 'Pre Req File' fields filled out. The 'Pre Req File Password' field contains '03bf4<R1[otb]641'. The 'Pre Req File' field has a 'Choose File' button and a 'Selected File: pre-req(Web).xlsx' label. There is a 'Submit' button. The 'Metadata' section is still visible, showing the same data as in Fig 13.5.

Fig 13.7: Fixed Pre-req uploading page

CHAPTER 7

REFLECTION AND ANALYSIS

The internship at CyRAACS provided a holistic exposure to modern software development practices within the cybersecurity domain. It was not just an opportunity to learn technologies but also to understand how secure systems are built and maintained in real-world scenarios. From contributing to internal tools like the VAPT Reporting Portal to building independent mini-projects, the journey offered a well-rounded experience of both technical development and professional growth. This reflection outlines the technical advancements, practical learnings, and professional maturity achieved through the internship.

During this period, I built a solid foundation in FastAPI, learning how to structure APIs, handle requests/responses, and connect to MongoDB Atlas for data operations. On the frontend, I practiced React.js fundamentals like state management, side effects, and context sharing, which helped me create dynamic interfaces and contribute to portal enhancements. To apply my learning, I created mini-projects including a To-Do List, Stopwatch, and Notes App. These projects allowed me to understand the integration of frontend and backend services, data validation, and file handling. The highlight was contributing to the VAPT Reporting Portal, where I worked on adding new features, improving UI, fixing bugs, and managing API-data flow efficiently.

I gained valuable exposure to real-world software development, version control using Git, and working collaboratively in a professional setting. Additionally, I learned about cybersecurity practices, such as vulnerability tracking, CVSS scoring, and how tools like Compass and VAPR together support a complete GRC solution.

In Conclusion, this internship gave me a strong foundation in full-stack development and introduced me to the role of cybersecurity in product development. It helped me grow technically, adapt to real-time challenges, and become more confident in developing secure and reliable software solutions. Overall, the experience has been both insightful and career-shaping.

CHAPTER 8

RECOMMENDATIONS

1. Introduce Well-Defined Internship Structures

CyRAACS can enhance future internship experiences by adopting a structured framework that sets clear expectations from the start. Defining roles, goals, timelines, and deliverables will help interns stay focused and aligned. A balanced schedule with interactive sessions and task-based learning can guide skill growth while enabling interns to track their development more effectively.

2. Foster Academic Collaborations

Establishing partnerships with universities and institutions can enrich internships by connecting theory with practical applications. Collaborations through workshops, guest lectures, or joint research activities can introduce interns to real-world challenges in cybersecurity, risk management, and compliance, making their learning more well-rounded.

3. Establish Strong Mentorship Channels

Assigning each intern a mentor who offers technical guidance and regular feedback can make the learning curve smoother. Structured mentor–intern check-ins, feedback reviews, and support can help interns better understand workflows and boost their confidence in applying tools and solving problems independently.

4. Encourage Active Participation and Creative Problem Solving

Creating opportunities for interns to contribute ideas, participate in meetings, and take ownership of smaller projects encourages innovation. This involvement not only enhances their learning but also brings fresh ideas to ongoing work, fostering a two-way benefit between the intern and the organization.

5. Create Networking and Exposure Opportunities

Facilitating cross-functional interactions can help interns explore various career paths. Organizing Q&A sessions, informal chats with professionals, or internal demo days allows interns to learn beyond their immediate teams, increasing awareness of the broader tech and cybersecurity landscape.

CHAPTER 9

CONCLUSION

My internship journey at CyRAACS has been an incredibly enriching and transformative experience. Over the course of this internship, I not only gained technical knowledge but also developed a deeper understanding of how real-world projects work in a professional environment. From learning FastAPI and MongoDB on the backend to applying core React concepts on the frontend, I got the chance to work on both sides of application development. This full-stack exposure helped me understand how different technologies come together to build real, functional software.

One of the highlights of my experience was building mini-projects like the To-Do List, Stopwatch, and Notes App, which gave me hands-on practice with state management, handling user input, and API communication. Later, working on full-stack applications involving FastAPI, MongoDB, and React helped me understand how to manage data flow, error handling, and validation in real-time scenarios. I also learned how to work with APIs, upload files securely, and follow a clean project structure.

Beyond technical skills, this internship helped me grow professionally. I learned the importance of collaboration, debugging effectively, and managing time while handling multiple tasks. Contributing to real projects, such as enhancing features on the portal, made me realize the value of writing efficient and clean code. I also got insights into cybersecurity practices, VAPT (Vulnerability Assessment and Penetration Testing), and GRC (Governance, Risk, and Compliance), which opened up my interest in how companies secure their digital assets.

Moreover, I got to see how mentorship and guidance from experienced team members can make a huge difference. Regular feedback and review sessions pushed me to think critically and improve with every task. The internship not only gave me technical confidence but also helped me understand the importance of communication, problem-solving, and team spirit in a work setting.

In conclusion, my time at CyRAACS has been a valuable stepping stone in my career journey. It has boosted my confidence as a developer, given me practical skills, and taught me how to adapt, learn, and contribute in a fast-paced tech environment. I am truly grateful for this experience and excited to apply what I've learned to future opportunities.

CHAPTER 10

REFERENCES

1. A. Asokan, "Automation Improvement in Cyber Risk Management," 2023 *International Conference on Machine Learning, Big Data, Cloud and Parallel Computing (COMITCon)*, Greater Noida, India, 2023, pp. 136-141, doi: [10.1109/COMITCon58584.2023.10271658](https://doi.org/10.1109/COMITCon58584.2023.10271658).
2. M. Garcia, "Concerns about Cybersecurity: The Implications of the use of ICT for Citizens and Companies," *International Journal of Arts, Design & Technology*, vol. 7, no. 1, 2024. doi: [10.55267/iadt.07.13226](https://doi.org/10.55267/iadt.07.13226).
3. T. Ahmad et al., "AI-Driven Cybersecurity: An Overview, Security Intelligence Modeling and Research Directions," *Preprints*, 2021. [Online]. Available: <https://www.preprints.org/manuscript/202101.0457/v1>.
4. M. Zharikova, "Analysis of the Market Conditions for Cybersecurity Products and Services," *Business Navigator*, no. 78, 2025. [Online]. Available: http://business-navigator.ks.ua/journals/2025/78_2025/76.
5. M. Ali et al., "Recent Advances in Cybersecurity and Safety Architectures in Automotive, IT, and Connected Services," *Journal of Universal Computer Science*, vol. 27, no. 10, pp. 1031–1052, 2021. doi: 10.3897/jucs.72072.
6. R. Rao et al., "Governance, Risk Management, and Compliance," 2024 *IEEE Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)*, pp. 214-220, doi: [10.1109/TPS-ISA59464.2024.10494630](https://doi.org/10.1109/TPS-ISA59464.2024.10494630).
7. R. M. Kumar et al., "Efficient Vulnerability Assessment and Penetration Testing: A Framework for Automation," 2023 *International Conference on Intelligent Computing and Control Systems (ICICCS)*, Madurai, India, 2023, pp. 1046-1051, doi: [10.1109/ICICCS58268.2023.10183397](https://doi.org/10.1109/ICICCS58268.2023.10183397).
8. N. Patel and V. K. Pandey, "Impact of Vulnerability Assessment and Penetration Testing (VAPT) on Operating System Security," 2023 *7th International Conference on Computing Methodologies and Communication (ICCMC)*, Erode, India, 2023, pp. 789-793, doi: [10.1109/ICCMC57384.2023.10094095](https://doi.org/10.1109/ICCMC57384.2023.10094095).
9. A. Singh and A. Sharma, "An Analysis of Vulnerability Scanners in Web Applications for VAPT," 2022 *6th International Conference on Computing Methodologies and Communication (ICCMC)*, Erode, India, 2022, pp. 1204-1209, doi: [10.1109/ICCMC53470.2022.9844365](https://doi.org/10.1109/ICCMC53470.2022.9844365).